



In brief. What each solution category can enforce, and on whose standards. Page 1 sets out why ownership matters; page 2 maps current vendor coverage; page 3 gives the operating pattern and the five contract tests for your next platform decision.

Purpose

Assessed against the *Agentic AI Capability Stack™*, Version 3.1. The matrix scores the seven capabilities where vendor coverage diverges, Layers 3 to 9; Layers 1 to 2 are commodity data infrastructure and Layers 10 to 11 are board mandate, neither vendor-differentiated. Vendors named are established examples, not a comprehensive market survey; inclusion does not constitute endorsement. Reviewed quarterly.

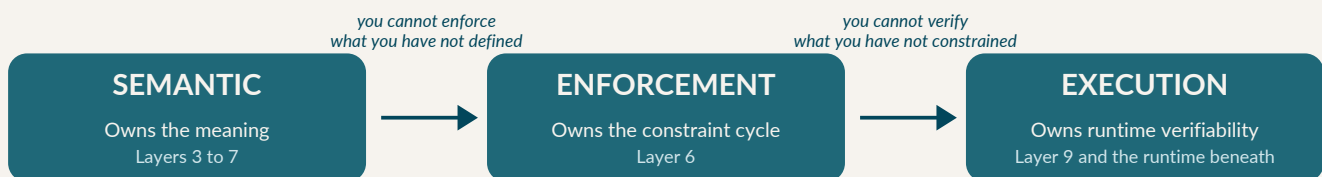
The eleven capabilities

Cap.	Name	What it does
11	Strategy	The board mandate and the investment case for AI
10	Operating Model	Who decides what: human and agent roles, decision rights, escalation
9	Agent Capability	The agents themselves: how they run, coordinate, and act
8	Agent Connectivity	One common way for agents to reach tools and governance (MCP, A2A)
7	Provenance	An auditable record of which data, rules, and authority produced each action (PROV-O)
6	Guardrails	Rules checked before an agent acts, not after the fact (SHACL)
5	Agent Contracts	What each agent is permitted to do, stated so machines can enforce it (ODRL)
4	Knowledge Graph	The connected, queryable record of the organisation's knowledge (RDF, SPARQL)
3	Ontology	The agreed meaning of the organisation's terms, so every agent reads them the same way (OWL, SKOS)
2	Data Connectivity	The pipelines and event streams that move data into the stack
1	Data	The organisation's data and authored knowledge, structured and unstructured

Capabilities 1–2 and 10–11 are not vendor-differentiated: data foundation is commodity infrastructure; strategy and operating model are board mandate. Capabilities 3 through 9 are the governance and capability core where coverage diverges and ownership determines sovereignty.

Why ownership matters

Three sovereignties, one chain. Each link is the precondition for the next.



Lose any link and it collapses to whoever owns the missing one, almost always the vendor.

The chain is a procurement test, not an architecture diagram. Semantic sovereignty asks whether the organisation, not the vendor, owns the meaning its agents reason over. Enforcement sovereignty asks whether a change in regulation can be applied on the organisation's own schedule or only when the vendor ships a release. Execution sovereignty asks whether the organisation can verify, independently of the provider, that an agent did what it was told. Each is a question about who holds the layer when the vendor relationship changes, and each points to a different requirement when technology is chosen.

None of this is an argument against proprietary technology. Proprietary engines and platform-specific formats have a legitimate place as derived execution caches, regenerated from an authoritative source the organisation holds on open standards. The question is only which copy is authoritative: rent that one, and every system downstream inherits the vendor's terms. Page 3 sets out the pattern.

Nor are these standards academic. Siemens runs its Industrial Knowledge Graph on them in production manufacturing. IKEA's knowledge graph drives recommendation and search from an ontology and controlled vocabularies maintained by domain experts. The BBC has published its ontologies openly and served Sport and News from them since 2010. Netflix built its Unified Data Architecture, published in 2025, on the same open standards, including the guardrail capability: production deployments at some of the most demanding engineering organisations in the world.

Based on publicly available information, Q2 2026. Not independent audit. Licensed CC BY-ND 4.0.

Cap.	Capability	Open Agent Runtime	Hyperscaler			Enterprise Ecosystem		Knowledge Graph Platforms	
		OpenClaw, NemoClaw	AWS Neptune + Bedrock	Microsoft Fabric IQ, Copilot Studio	Google Vertex AI, Spanner Graph	SAP Joule, AI Agent Hub	ServiceNow AI Control Tower	Stardog, Graphwise, TopQuadrant, eccenca, metaphacts	Neo4j
9	Agent Capability	■ no governance	● Bedrock	● Copilot Studio	● Vertex AI	■ Joule	■ AI agents	— (emerging)	—
8	Agent Connectivity	● MCP/A2A	■ MCP	■ MCP	● A2A	■ wrapped	■ Action Fabric	● MCP	—
7	Provenance	—	■ AgentCore	■ Purview	■ Agent ID	■ Agent Hub	■ Control Tower	● PROV-O	—
6	Guardrails	—	■ AgentCore	■ Content Safety	■ Gateway	■ Joule Studio	■ Control Tower	● SHACL	■ n10s post-hoc
5	Agent Contracts	—	■ AgentCore	■ Copilot Ctrl	■ Gateway	■ Agent Hub	■ Veza / scoped	● ODRL/OWL	—
4	Knowledge Graph	—	● Neptune	■ Fabric IQ (prev)	■ Spanner	■ BDC	■ CMDB	● SPARQL	■ Cypher
3	Ontology	—	● Neptune	■ Fabric IQ (prev)	■ Spanner Graph	■ SAP KG	■ CSDM	● OWL/RDF	■ n10s (RDF import)

● On standards the organisation can own ■ Present but vendor-owned — Absent at the level required (preview) = not yet generally available

Key finding

The dedicated knowledge-graph platforms (**Stardog, Graphwise, TopQuadrant, eccenca, metaphacts**) can cover Layers 3 through 8 on open standards the organisation owns. No other solution category provides equivalent coverage on portable standards. Hyperscalers and enterprise platforms provide functional governance at L3–L7, but on proprietary formats that cannot be exported or federated independently. Open agent runtimes cover L8–L9 but are absent at L3–L7 entirely.

The procurement question is not which vendor covers the most layers. It is which layers the organisation must own on standards it controls. The meaning architecture (L3), the knowledge graph (L4), the agent contracts (L5), the guardrails (L6), and the provenance trail (L7) encode institutional knowledge and governance logic. Placing these on vendor-owned formats means rebuilding on every platform change.

Vendor notes

Microsoft. Fabric IQ, announced at Ignite in November 2025, is the hyperscaler investment most aligned with the Stack's meaning layer (L3–L4). Six months on, it remains in preview with no published general-availability date. The gap is not ignorance; it is difficulty. Building a production-grade meaning layer is hard, which is precisely why the CDO mandate exists. As of May 2026 the format Fabric IQ stores its meaning in is undocumented, and there is no open way to query or export it; whether an organisation can own and move that layer independently of Microsoft remains open until Microsoft ships portability.

SAP and ServiceNow. Both expanded governance offerings in May 2026. SAP introduced the AI Agent Hub on LeanIX at Sapphire 2026 (general availability Q3 2026, free with the Business AI Platform); ServiceNow expanded AI Control Tower at Knowledge 2026 with NVIDIA and Microsoft Agent 365 integrations. Coverage is functionally dense at L5–L7 in both cases, but the underlying meaning, on both sides, stays on vendor-owned stores.

Amazon. Neptune Analytics can import open-standard graphs but stores and runs them in its own property-graph form, so the version that actually operates is vendor-defined. Convergence at the surface does not change the ownership question underneath.

Neo4j and property-graph platforms. A property graph is a graph, not by itself a meaning layer. It stores entities and relationships with local identifiers and enforces basic constraints, but the formal definitions, rules, and provenance that let meaning be validated and carried between systems are added in application code rather than authored as portable, open-standard artefacts. These platforms can be extended toward semantics and are strong operational graphs; the ownership question is whether the meaning is the organisation's to move, or the vendor's to hold.

Palantir. Managed deployment platforms (Palantir AIP with Forward Deployed Engineers) can cover L3–L9 comprehensively as a proprietary service. They sit outside this diagnostic because their delivery model differs structurally from software platforms an organisation operates itself.

Watch signals for Q3

RDF 1.2. A new version of the core open standard reached a near-final stage in April 2026. The signal to watch is vendor support: once two or more major graph engines ship it, the main remaining technical objection to running open standards for live workloads falls away, and any decision to commit to a single vendor's proprietary graph format deserves revisiting.

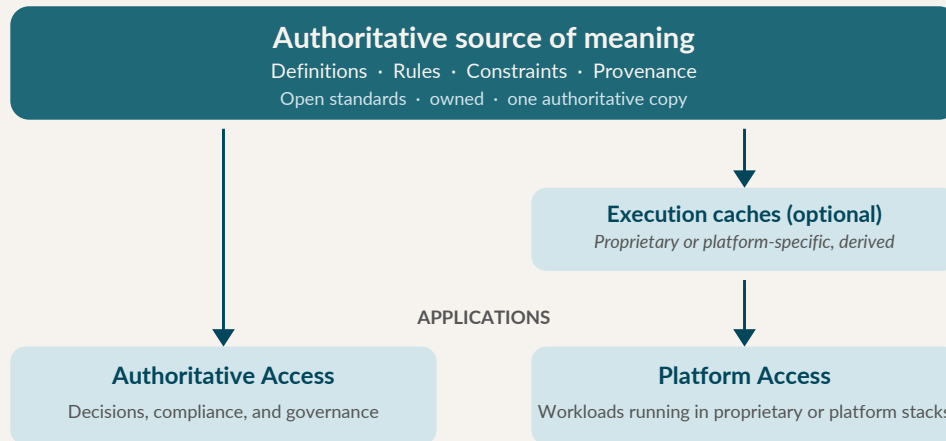
Open Semantic Interchange. ServiceNow joined the OSI in June 2026. It moves no cell here: the Diagnostic scores shipped coverage and ownership, not membership, and OSI standardises how metrics and definitions are exchanged, not the reasoning, rule-checking, and audit the governance layers depend on; ServiceNow's own meaning stays on vendor-owned stores, so the amber marks at L3 to L5 hold. The signal worth watching sits upstream of any one vendor: OSI has chartered a working group to map its concepts onto formal meaning standards, and whether that lands on the open standards in this stack or on a simpler format without formal meaning decides whether OSI becomes a bridge to the open layer or a parallel format beside it. A mapping onto the open standards would strengthen the case made here. ServiceNow membership matters to the mapping only once it ships compliant import and export of meaning models, comparable in weight to the RDF 1.2 signal above.

Canonical reference: The Agentic AI Capability Stack™, Version 3.1. Based on publicly available information, Q2 2026. Not independent audit. The Diagnostic assesses coverage and ownership on open standards, not performance; vendor performance benchmarks should be validated against organisational workloads. Licensed CC BY-ND 4.0 with attribution.



One authoritative source, two access paths

Authority stays in the meaning layer. Everything else is derived from it.



The access path is a choice of risk and fit, not a limit of the source.

The recommended pattern keeps one authoritative source and derives everything else from it. The organisation holds the canonical definition of its meaning, its rules, and its audit trail on open standards it owns. Where a proprietary or platform-specific stack already runs the work, the organisation derives an execution cache from that authoritative source rather than authoring the meaning a second time. Applications then choose how they read: directly from the meaning layer for decisions that carry risk, compliance, or governance weight, and through a derived cache where the work runs inside a proprietary or platform-specific stack.

This makes the access path a question of risk and fit. A derived cache earns its place when the organisation can replace it, or change the proprietary stack beneath it, and regenerate an equivalent from the authoritative source without losing meaning, rules, or audit. Authority never moves into the derived cache. That single line, what is authoritative and what is derived, separates using a proprietary stack from depending on one.

The pattern is in production. Netflix's Unified Data Architecture holds one authoritative model on open standards and projects it into every representation its platforms need; its own name for the principle is “model once, represent everywhere”.

The coverage matrix on page 2 reads directly through this pattern. A green mark identifies a store that can hold the authoritative copy, on standards the organisation owns. An amber mark identifies a legitimate execution cache: usable, often valuable, never the place the meaning is defined. The matrix is therefore a map of what may be cached and what must be owned.

Five contract tests

The pattern becomes enforceable when it is written into vendor contracts. Articles 3a and 3b of the series derive five criteria that function as deal-stoppers in platform evaluation:

Test	What the contract must guarantee	Sovereignty protected
1. Configurability	Governance rules are client-configurable without a vendor software release	Enforcement
2. Enforcement timing	Runtime enforcement fires before execution, not as logging after the fact	Enforcement
3. Portability	The meaning layer sits on open standards, portable across platforms	Semantic
4. Provenance	Every agent decision links, at the moment of action, to the governance framework in force at that time	Semantic
5. Verifiable execution	The runtime produces evidence the agent ran as specified, independent of the infrastructure provider's word	Execution

Any one absent, and the corresponding sovereignty rests with the vendor regardless of the organisational chart.

Where common terms sit

MCP and **A2A** are connectivity protocols at Layer 8: they standardise how agents reach tools and governance, not what the governance says. **GraphRAG** is a retrieval pattern that reads from the knowledge graph (Layer 4) to improve answers; it consumes the meaning layer and is not a substitute for owning it. **BI semantic layers** define metrics and dimensions at the analytics boundary; they sit above the stack and are not the meaning layer (Layer 3), which carries the formal definitions agents reason and are governed against.